Jin Sheng Cao

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EDUCATION

University of Maryland College Park, MD
B.S., Mechanical Engineering, GPA: 3.83 Expected May 2029
Coursework: Statics, Dynamics, Mechanics of Materials, Engineering Design, CAD (SolidWorks), Python & MATLAB, Diff EQ
- QUEST (Quality Enhancement Systems and Teams) Honors Program - Consultant (Cohort 48)

SKILLS

CAD & Analysis: SolidWorks (CSWA), Onshape, Fusion 360, Siemens NX (learning), Ansys Mechanical (FEA), Stress & Thermal Analysis, Factor of Safety, GD&T (ASME Y14.5), DFM, Tolerance Analysis, Hand Calculations, FRACAS
Manufacturing & Test: Mechanisms, Actuators, 3D Printing, Rapid Prototyping, Sheet Metal, Verification Testing, RCCA
Programming & Systems: Python, MATLAB, C++, Java, Arduino, Excel, PowerPoint, BOMs, Design Reviews, Teamwork

WORK EXPERIENCE

UMD Bioinspired Robotics Laboratory **January 2026 – Present**
Undergraduate Student Researcher College Park, MD
- Design, 3D-printed, and delivered a working bio-inspired autonomous underwater vehicle (AUV) in **2 weeks across 2 iterations**, modeling the full vehicle in SolidWorks with a custom tail and fins for propulsion and steering.
- Engineer a split-hull design with internal mounts to integrate and package environmental sensors, motors, and electronics within a 2x2x2 ft base, meeting volume constraints and **optimizing part size to cut print and post-processing time by 5%.**
- Plan and ran sealing, structural, and buoyancy tests (15+ s upright float test); led RCCA on leaks traced to poor interlayer adhesion; redesigned hull CAD with an O-ring seal targeting the root cause and testing for a fully waterproof hull.

Johns Hopkins University Applied Physics Laboratory **August 2024 – June 2025**
Optics and Photonics Intern Laurel, MD
- Developed Python automation tool that **generated and analyzed 10,000+ data points** to predict optical parametric oscillator (OPO) configurations, cutting system tuning time from hours to under a minute **(95%+ reduction)**
- Automated real-time signal-peak detection with an optical spectrum analyzer, cutting hours of manual searching to minutes.
- Analyzed experimental OPO data in MATLAB to find problems within the machine and identify failure modes and performance trends, informing improvements and where to add automation in infrared laser hardware.

ENGINEERING PROJECTS

UMD Robotics at Maryland - Mars Rover Team **October 2026 – Present**
Lead Mechanical Engineer College Park, MD
- **Lead 20+ engineer team** through requirements, PDR, design, fabrication, and test, taking a 6-DOF arm and science payload from zero to a successful competition demo in one year; **present design reviews to 50+ students, faculty, and industry.**
- Design CF-nylon arm linkages with stepper-driven joints, bearings, encoders, and a 1:4 gear train servo gripper using DFM; produce GD&T drawings and BOMs; fabricate via 3D printing, waterjet, and sheet metal manufacturing.
- Establish Ansys FEA workflow for the team, validating 3 structural and thermal load cases with hand calculations; redesign a failing crank (SF 0.53) to SF 3.3, **cutting raw material cost 15% and confirm gripper holds 5 kg max payload at SF 3.7.**

UMD TERP RAPTOR **September 2026 – Present**
Build and Integration Engineer College Park, MD
- Own 3D integrated CubeSat model as Responsible Engineer for PDR with partners including **The Aerospace Corporation**; designed in SolidWorks, 3D printed, and assembled with custom brackets and joints to verify fit and assembly order.
- Responsible for upcoming flight build and integration of UMD's first CubeSat with NASA GSFC; plan assembly sequences with the 3D-printed model, verify fit, and flag build errors early to protect flight hardware and keep the build on schedule.
- Author manufacturing and integration procedures for the Assembly, Integration and Test (AI&T) campaign; develop verification test plan, cleanroom preparation, and test reports for upcoming thermal vacuum (TVAC) and vibration qualification.

Team V.A.A.C.E. (Venus Aerobot for Atmosphere & Cloud Exploration) **August 2026 – Present**
Mechanical & Assembly Engineer College Park, MD
- Owned the design and assembly of the Y-axis subsystem of a robotic skin system in Fusion 360 for a **$20K NASA L'SPACE seed-funded** Venus aerobot mission, finishing 5 days ahead of deadline and enabling successful subsystem prototype testing.
- Unify sub-assemblies from a 20-person team into the full aerobot CAD assembly, guiding teammates through design fixes and resolving build errors; verifying the final assembly, clearing it for Ansys Mechanical analysis and NASA L'SPACE reviews.

LEADERSHIP

IEEE @ UMD **October 2026 – Present**
Corporate Committee Marketing Chair College Park, MD
- Lead marketing by email for industry and networking events, with over **300 RSVP and attendees** for UMD engineering events.
- Conduct school-wide email campaigns to boost engineering student engagement and cold-email corporate recruiters to host networking events, strengthening UMD's industry sponsor partnerships.